An Introduction to Ergodicity Economics

Ole Peters and Alexander Adamou

Long-term growth rates

Postulate that agents maximize wealth.
Twist: not expected wealth but real wealth over time.

Ergodicity is broken
$\rightarrow$ expected wealth $\neq$ real wealth over time.

1.1, p.2: The game

- The infamous coin toss

1.1, p.2: The game

- The infamous coin toss
  - p.2

Let us define our model of the dynamics of your wealth under the rules we just specified.[1] At regular intervals of duration δt, we randomly generate a factor r with each possible value $r_i \in \{0.6, 1.5\}$ occurring with probability $1/2$,

$$r = \begin{cases} 0.6 & \text{with probability } 1/2 \\ 1.5 & \text{with probability } 1/2 \end{cases} \tag{1.1}$$

and multiply current wealth by that factor, so that

$$x(t + \delta t) = x(t)r. \tag{1.2}$$

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  - discrete form of Geometric Brownian Motion (GBM): applies to all of finance theory.
  - multiplicativity: evolution, life, investing, learning, growing.

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- The leveraged coin toss: only gamble with fraction l of your wealth.

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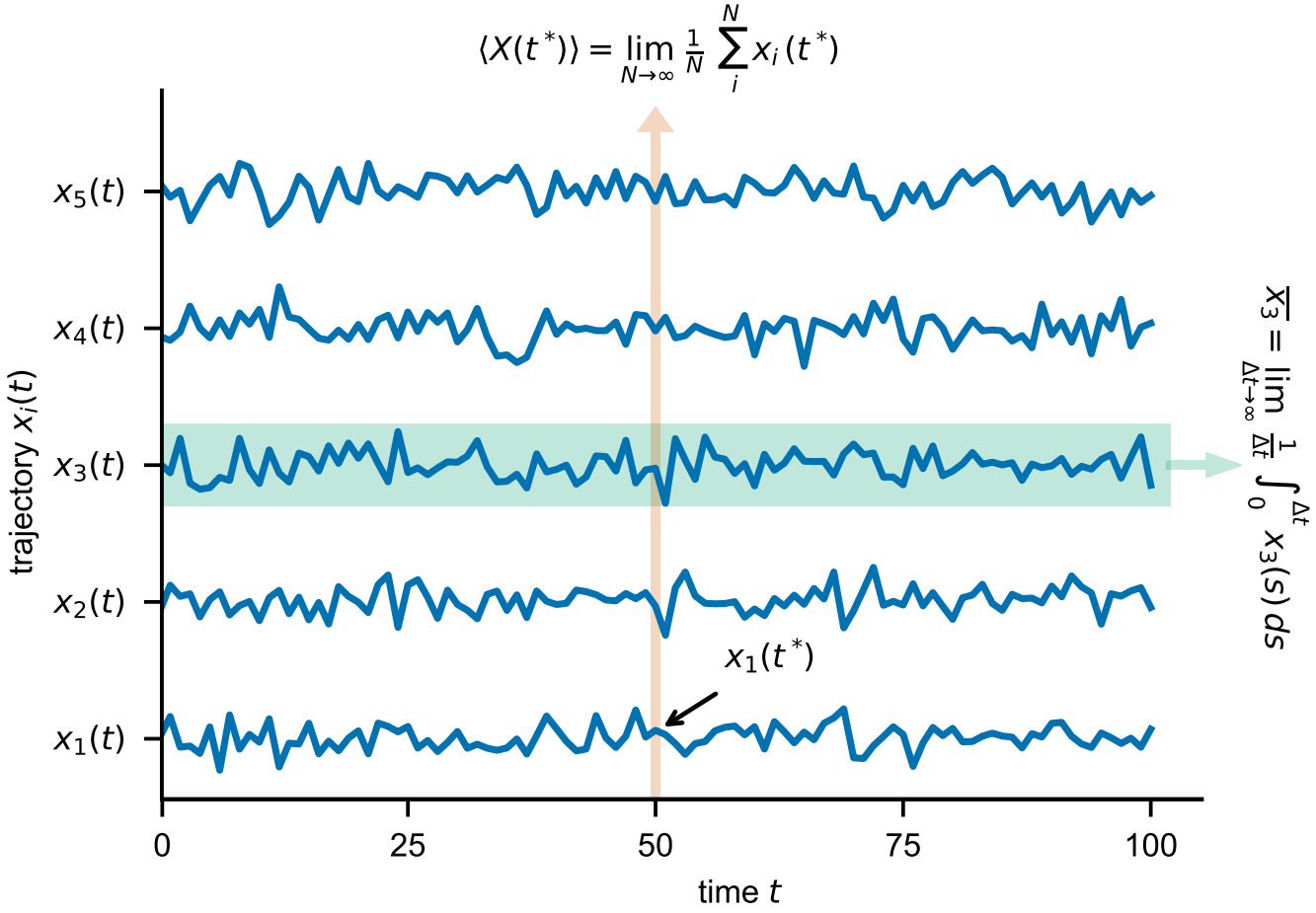
- The cooperating coin toss: pool and share with others.

1.1, p.2: The game

- The infamous coin toss
- The leveraged coin toss: only gamble with fraction l of your wealth.
- The cooperating coin toss: pool and share with others.
- The social coin toss: pool and share a bit with others.

Chapter 2, p.9: Averages



Chapter 3, p.25: Ergodicity

Richard H Thaler
@R_Thaler

This is NNT hogwash.

Philosophy: keep the text light.
Relevant ergodic problem: expected value $\neq$ time average.
But there's 150 years of detailed theory behind this.

Ergodic hierarchy

Ergodicity means

- Metric transitivity

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Ergodicity means

- Metric transitivity
  Dynamical sytem: (X, F, μ, T)
  For any sets $A, B \in F$, there exists an n such that $\mu(T^{-n}(A) \cap B) > 0$.
  It follows that if A satisfies $T^{-1}(A) = A$, then either $\mu(A) = 1$ or $\mu(A) = 0$.

Ergodic hierarchy

Ergodicity means

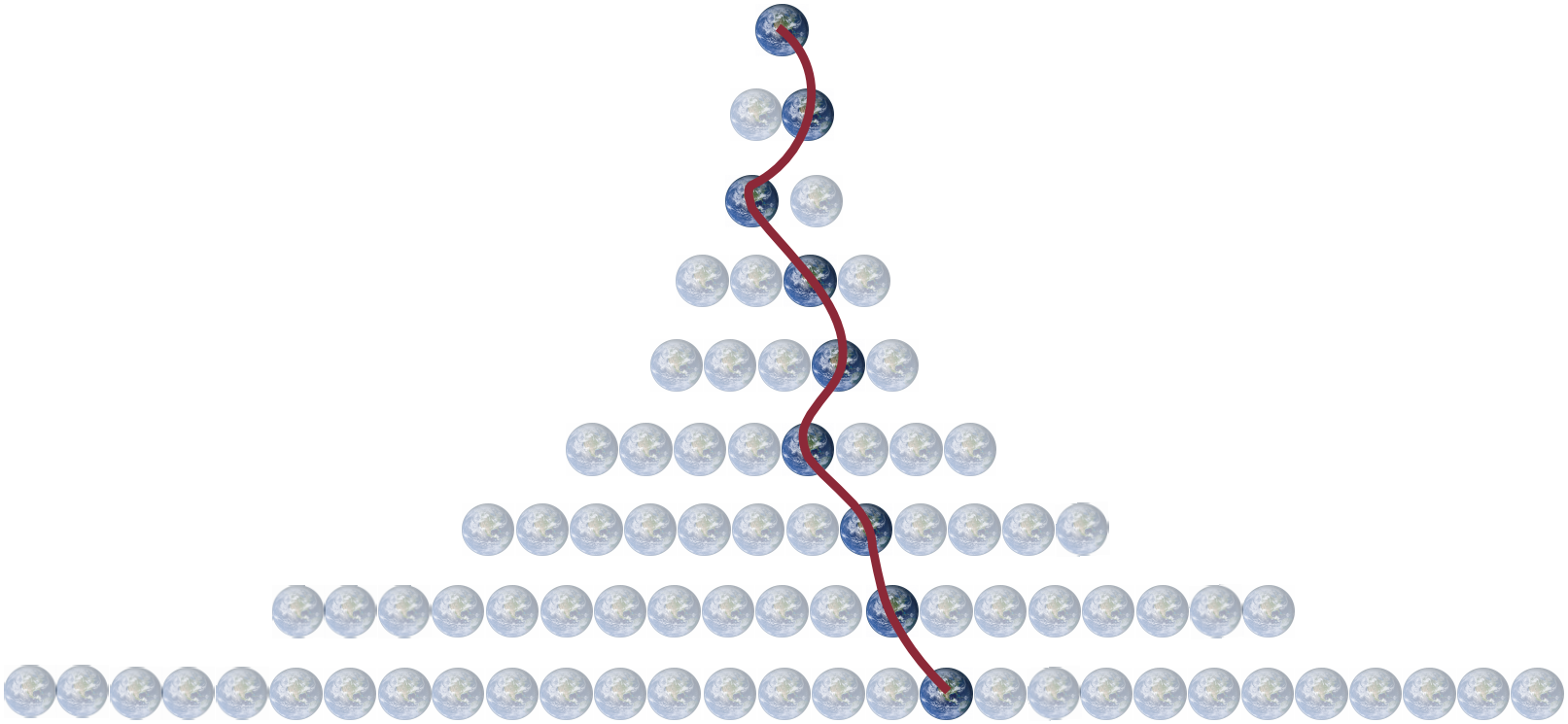
- Metric transitivity

$\rightarrow$ Equality of distributions

$\rightarrow$ Equality of averages

Chapter 4, p.34: Models

More tools.

- Brownian motion
- Stochastic differential equations
- Itôcalculus
- Geometric Brownian motion

Thank you.